1.

```java
package com.payoda.training.ooc;
class Student {
	protected String name;
	protected String id;
	protected int age;
	protected double grade;
	protected String address;
	public Student() {
		this.name = "";
		this.id = "";
		this.age = 0;
		this.grade = 0.0;
		this.address = "";
	}
	public Student(String name, String id, int age, double grade, String address) {
		this.name = name;
		this.id = id;
		this.age = age;
		this.grade = grade;
		this.address = address;
	}
	public String getName() {
		return name;
	}
	public void setName(String name) {
		this.name = name;
	}
	public String getId() {
		return id;
	}
	public void setId(String id) {
		this.id = id;
	}
	public int getAge() {
		return age;
	}
	public void setAge(int age) {
		this.age = age;
	}
	public double getGrade() {
		return grade;
	}
	public void setGrade(double grade) {
		this.grade = grade;
	}
	public String getAddress() {
```

```java
		return address;
	}
	public void setAddress(String address) {
		this.address = address;
	}
	public void display() {
		System.out.println("Name: " + name);
		System.out.println("ID: " + id);
		System.out.println("Age: " + age);
		System.out.println("Grade: " + grade);
		System.out.println("Address: " + address);
	}
	public boolean isPassed() {
		return grade > 50;
	}
}
class UGStudent extends Student {
	private String degree;
	private String stream;
	public UGStudent() {
		super();
		this.degree = "";
		this.stream = "";
	}
	public UGStudent(String name, String id, int age, double grade, String address, String degree,
String stream) {
		super(name, id, age, grade, address);
		this.degree = degree;
		this.stream = stream;
	}
	public String getDegree() {
		return degree;
	}
	public void setDegree(String degree) {
		this.degree = degree;
	}
	public String getStream() {
		return stream;
	}
	public void setStream(String stream) {
		this.stream = stream;
	}
	@Override
	public void display() {
		System.out.println("UG Student Details:");
		System.out.println("Name: " + name);
		System.out.println("ID: " + id);
		System.out.println("Age: " + age);
		System.out.println("Grade: " + grade);
```

```java
            System.out.println("Address: " + address);
            System.out.println("Degree: " + degree);
            System.out.println("Stream: " + stream);
    }
    @Override
    public boolean isPassed() {
            return grade > 70;
    }
}
class PGStudent extends Student {
    private String specialization;
    private int noOfPapersPublished;
    public PGStudent() {
            super();
            this.specialization = "";
            this.noOfPapersPublished = 0;
    }
    public PGStudent(String name, String id, int age, double grade, String address, String
specialization,
                    int noOfPapersPublished) {
            super(name, id, age, grade, address);
            this.specialization = specialization;
            this.noOfPapersPublished = noOfPapersPublished;
    }
    public String getSpecialization() {
            return specialization;
    }
    public void setSpecialization(String specialization) {
            this.specialization = specialization;
    }
    public int getNoOfPapersPublished() {
            return noOfPapersPublished;
    }
    public void setNoOfPapersPublished(int noOfPapersPublished) {
            this.noOfPapersPublished = noOfPapersPublished;
    }
    @Override
    public void display() {
            System.out.println("PG Student Details:");
            System.out.println("Name: " + name);
            System.out.println("ID: " + id);
            System.out.println("Age: " + age);
            System.out.println("Grade: " + grade);
            System.out.println("Address: " + address);
            System.out.println("Specialization: " + specialization);
            System.out.println("Number of Papers Published: " + noOfPapersPublished);
    }
    @Override
    public boolean isPassed() {
```

```
                return grade > 70 && noOfPapersPublished >= 2;
        }
}
public class Main {
        public static void main(String[] args) {
                Student s1 = new Student("Dharanesh", "S100", 21, 58, "Erode");
                s1.display();
                System.out.println("Passed: " + s1.isPassed());
                System.out.println();
                UGStudent ug1 = new UGStudent("Surya", "UG101", 19, 72, "Coimbatore", "B.E",
"Electronics and Communication");
                ug1.display();
                System.out.println("Passed: " + ug1.isPassed());
                System.out.println();
                PGStudent pg1 = new PGStudent("Karthi", "PG102", 25, 78, "Tirupur", "Data Analytics",
4);
                pg1.display();
                System.out.println("Passed: " + pg1.isPassed());
                System.out.println();
        }
}
```

```
Name: Dharanesh
ID: S100
Age: 21
Grade: 58.0
Address: Erode
Passed: true

UG Student Details:
Name: Surya
ID: UG101
Age: 19
Grade: 72.0
Address: Coimbatore
Degree: B.E
Stream: Electronics and Communication
Passed: true

PG Student Details:
Name: Karthi
ID: PG102
Age: 25
Grade: 78.0
Address: Tirupur
Specialization: Data Analytics
Number of Papers Published: 4
Passed: true
```

2.

```
package com.payoda.training.ooc;
import java.util.Scanner;
class Vehicle {
        protected String make;
        protected String vehicleNumber;
        protected String fuelType;
        protected int fuelCapacity;
        protected int cc;
```

```java
	public Vehicle(String make, String vehicleNumber, String fuelType, int fuelCapacity, int cc) {
		this.make = make;
		this.vehicleNumber = vehicleNumber;
		this.fuelType = fuelType;
		this.fuelCapacity = fuelCapacity;
		this.cc = cc;
	}
	public String getMake() {
		return make;
	}
	public void setMake(String make) {
		this.make = make;
	}
	public String getVehicleNumber() {
		return vehicleNumber;
	}
	public void setVehicleNumber(String vehicleNumber) {
		this.vehicleNumber = vehicleNumber;
	}
	public String getFuelType() {
		return fuelType;
	}
	public void setFuelType(String fuelType) {
		this.fuelType = fuelType;
	}
	public int getFuelCapacity() {
		return fuelCapacity;
	}
	public void setFuelCapacity(int fuelCapacity) {
		this.fuelCapacity = fuelCapacity;
	}
	public int getCc() {
		return cc;
	}
	public void setCc(int cc) {
		this.cc = cc;
	}
	public void displayMake() {
		System.out.println("*** " + make + " ***");
	}
	public void displayBasicInfo() {
		System.out.println("--- Basic Information ---");
		System.out.println("Vehicle Number: " + vehicleNumber);
		System.out.println("Fuel Type: " + fuelType);
		System.out.println("Fuel Capacity: " + fuelCapacity);
		System.out.println("CC: " + cc);
	}
	public void displayDetailInfo() {
	}
```

```java
}
class TwoWheeler extends Vehicle {
	private boolean kickStartAvailable;
	public TwoWheeler(String make, String vehicleNumber, String fuelType, int fuelCapacity, int cc,
				boolean kickStartAvailable) {
		super(make, vehicleNumber, fuelType, fuelCapacity, cc);
		this.kickStartAvailable = kickStartAvailable;
	}
	public boolean isKickStartAvailable() {
		return kickStartAvailable;
	}
	public void setKickStartAvailable(boolean kickStartAvailable) {
		this.kickStartAvailable = kickStartAvailable;
	}
	@Override
	public void displayDetailInfo() {
		System.out.println("Kick Start Available: " + (kickStartAvailable ? "Yes" : "No"));
	}
}
class FourWheeler extends Vehicle {
	private String audioSystem;
	private int numberOfDoors;
	public FourWheeler(String make, String vehicleNumber, String fuelType, int fuelCapacity, int cc,
String audioSystem,
				int numberOfDoors) {
		super(make, vehicleNumber, fuelType, fuelCapacity, cc);
		this.audioSystem = audioSystem;
		this.numberOfDoors = numberOfDoors;
	}
	public String getAudioSystem() {
		return audioSystem;
	}
	public void setAudioSystem(String audioSystem) {
		this.audioSystem = audioSystem;
	}
	public int getNumberOfDoors() {
		return numberOfDoors;
	}
	public void setNumberOfDoors(int numberOfDoors) {
		this.numberOfDoors = numberOfDoors;
	}
	@Override
	public void displayDetailInfo() {
		System.out.println("Audio System: " + audioSystem);
		System.out.println("Number Of Doors: " + numberOfDoors);
	}
	public class program2 {
		public static void main(String[] args) {
			Scanner sc = new Scanner(System.in);
```

```
		System.out.println("1. Two Wheeler\n2. Four Wheeler");
		System.out.print("Enter Vehicle Type: ");
		int choice = sc.nextInt();
		sc.nextLine();
		if (choice == 1) {
			System.out.print("Enter Make: ");
			String make = sc.nextLine();
			System.out.print("Enter Vehicle Number: ");
			String vehicleNumber = sc.nextLine();
			System.out.print("Enter Fuel Type: ");
			String fuelType = sc.nextLine();
			System.out.print("Enter Fuel Capacity: ");
			int fuelCapacity = sc.nextInt();
			System.out.print("Enter CC: ");
			int cc = sc.nextInt();
			sc.nextLine();
			System.out.print("Kick Start Available (true/false): ");
			boolean ksa = sc.nextBoolean();
			TwoWheeler tw = new TwoWheeler(make, vehicleNumber, fuelType,
fuelCapacity, cc, ksa);
			tw.displayMake();
			tw.displayBasicInfo();
			tw.displayDetailInfo();
		} else if (choice == 2) {
			System.out.print("Enter Make: ");
			String make = sc.nextLine();
			System.out.print("Enter Vehicle Number: ");
			String vehicleNumber = sc.nextLine();
			System.out.print("Enter Fuel Type: ");
			String fuelType = sc.nextLine();
			System.out.print("Enter Fuel Capacity: ");
			int fuelCapacity = sc.nextInt();
			System.out.print("Enter CC: ");
			int cc = sc.nextInt();
			sc.nextLine();
			System.out.print("Audio System: ");
			String audioSystem = sc.nextLine();
			System.out.print("Number Of Doors: ");
			int nod = sc.nextInt();
			FourWheeler fw = new FourWheeler(make, vehicleNumber, fuelType,
fuelCapacity, cc, audioSystem, nod);
			fw.displayMake();
			fw.displayBasicInfo();
			fw.displayDetailInfo();
		} else {
			System.out.println("Invalid choice");
		}
		sc.close();
	}
```

```
	}
}
```

```
1. Two Wheeler
2. Four Wheeler
Enter Vehicle Type: 1
Enter Make: yamaha
Enter Vehicle Number: 9181
Enter Fuel Type: petrol
Enter Fuel Capacity: 12
Enter CC: 150
Kick Start Available (true/false): false
*** yamaha ***
--- Basic Information ---
Vehicle Number: 9181
Fuel Type: petrol
Fuel Capacity: 12
CC: 150
Kick Start Available: No
```

3.

```
package com.payoda.training.ooc;
//Main for Associate
import java.util.Scanner;
//Shape.java
class Shape {
	double calculateArea() {
		return 0;
	}
}
//Square.java
class Square extends Shape {
	private double side;
	public Square(double side) {
		this.side = side;
	}
	@Override
	double calculateArea() {
		return side * side;
	}
}
//Rectangle.java
class Rectangle extends Shape {
	private double length, width;
	public Rectangle(double l, double w) {
		this.length = l;
		this.width = w;
	}
	@Override
	double calculateArea() {
		return length * width;
```

```java
	}
}
//Triangle.java
class Triangle extends Shape {
	private double base, height;
	public Triangle(double b, double h) {
		this.base = b;
		this.height = h;
	}
	@Override
	double calculateArea() {
		return 0.5 * base * height;
	}
}
public class program3 {
	public static void main(String args[]) {
		Shape s1 = new Square(5);
		Shape s2 = new Rectangle(3, 4);
		Shape s3 = new Triangle(4, 6);
		Shape[] shapes = { s1, s2, s3 };
		for (Shape s : shapes)
			System.out.println("Area: " + s.calculateArea());
	}
}
```

```
Area: 25.0
Area: 12.0
Area: 12.0
```

4.

```java
package com.payoda.training.ooc;
//Main for Associate
import java.util.Scanner;
class Associate {
	private int associateId;
	private String associateName;
	private String workStatus;
	public Associate() {
	}
	public Associate(int id, String name, String status) {
		this.associateId = id;
		this.associateName = name;
		this.workStatus = status;
	}
```

```java
	public int getAssociateId() {
		return associateId;
	}
	public void setAssociateId(int associateId) {
		this.associateId = associateId;
	}
	public String getAssociateName() {
		return associateName;
	}
	public void setAssociateName(String associateName) {
		this.associateName = associateName;
	}
	public String getWorkStatus() {
		return workStatus;
	}
	public void setWorkStatus(String workStatus) {
		this.workStatus = workStatus;
	}
	public void trackAssociateStatus(int numDays) {
		if (numDays <= 20)
			this.workStatus = "Core skills";
		else if (numDays <= 40)
			this.workStatus = "Advanced modules";
		else if (numDays <= 60)
			this.workStatus = "Project phase";
		else
			this.workStatus = "Deployed in project";
	}
}
public class program4 {
	public static void main(String[] args) {
		Associate associate = new Associate();
		Scanner sc = new Scanner(System.in);
		System.out.print("Enter Associate Id: ");
		associate.setAssociateId(sc.nextInt());
		sc.nextLine();
		System.out.print("Enter Associate Name: ");
		associate.setAssociateName(sc.nextLine());
		System.out.print("Enter number of days: ");
		int days = sc.nextInt();
		associate.trackAssociateStatus(days);
		System.out.println("Associate Id: " + associate.getAssociateId());
		System.out.println("Associate Name: " + associate.getAssociateName());
		System.out.println("Work Status: " + associate.getWorkStatus());
		sc.close();
	}
}
```

```
Enter Associate Id: 12
Enter Associate Name: dharanesh
Enter number of days: 200
Associate Id: 12
Associate Name: dharanesh
Work Status: Deployed in project
```